GITAM Deemed to be University

Information Technology | Course Code: IT303

Cyber Security & Cryptography - Cryptography Algorithms, Public Key Infrast

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Overview: Mathematical foundations: Modular arithmetic, Fermat Little Theorem, AES substitution

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Symmetric Cryptography & Block Ciphers
Feistel cipher structure, DES, Triple-DES, Advanced Encryption Standard AES rounds, Modes of operati

Unit 2: Asymmetric Cryptography & Key Exchange
Modular arithmetic, Extended Euclidean algorithm, RSA cryptosystem, Diffie-Hellman key exchange, Dis

Unit 3: Hash Functions & Message Authentication
Cryptographic hash properties, SHA-256, HMAC construction, Digital signatures DSA, ECDSA, Public Key

Unit 4: Network Protocols & Defenses
TLS 1.3 handshake, IPsec AH and ESP, Firewalls Stateful packet inspection, Intrusion Detection Syste

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Trace the mathematical steps of RSA algorithm: calculate public and private keys for primes p=17, q=11
2. Describe the Diffie-Hellman key exchange protocol and analyze its vulnerability to Man-In-The-Middle M
3. Formulate the four transformations applied in each round of the Advanced Encryption Standard AES ciphe

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* William Stallings - Cryptography and Network Security: Principles and Practice Pearson 7th Ed
* Bruce Schneier - Applied Cryptography: Protocols, Algorithms, and Source Code in C Wiley

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